
Methods of Calculation and Design for Mechanical and Public Health Building Services

How-To: Mechanical and Public Health Building Services
System Design

Author:

Adam Rees

Mechanical Building Services Calculation Methods

By

Adam Rees

©Adam Rees, 2019

Contents

Acronyms	1
I Ventilation	3
1 Introduction	4
1.1 Energy Balances	4
1.2 Units	5
1.3 Shadow Gaps and Door Undercuts	6
1.4 Ventilation Rates	7
1.4.1 Pollutant Control	8
1.4.2 Kitchen Ventilation Rates	8
2 Psychrometrics	11
2.1 Psychrometric Charts	11
2.1.1 Psychrometric Processes	11
2.1.2 ADP Calculations	11
2.1.3 Room Condition Calculations	11
2.1.4 Enthalpy Calculations	11
2.2 Manual Psychrometric Calculations	11
3 Heating and Cooling via Air	12
3.1 Heating Requirements	12
3.2 Cooling Requirements	12
3.3 Comfort and Efficiency	12
3.4 Calculations	12
3.4.1 Heating by Air	12
3.4.2 Cooling by Air	12
4 Ventilation Plant	13
4.1 Air Handling Unit (AHU) Components	13
4.1.1 Energy Recovery Devices	13
4.1.2 Energy Recovery Calculations	15
4.1.3 Frost Coils	16
4.1.4 Filtration	20
4.1.5 Fans	20

4.1.6 Heating/Cooling Coils	20
4.2 Component Resistance	20
4.3 Fan Heat Gain	20
4.4 Ventilation Plant Running Costs	20
5 Air Distribution	21
5.1 Displacement Ventilation	21
5.2 Ductwork Component Pressure Loss	21
II Heating and Cooling	22
6 Introduction	23
6.1 Steady State Heating	23
6.1.1 Conduction (Fabric) Heat Loss	23
6.1.2 Infiltration Heat Loss	24
6.1.3 Ventilation Heat Loss	24
6.1.4 Intermittent Correction Factor	24
6.1.5 Peak Heating Load	24
6.1.6 Distribution Losses	24
6.1.7 Infiltration Diversity Ratio	24
6.1.8 Central Plant Size	24
6.2 Annual Estimation of Heating Loads	24
III Domestic Water Services	25
7 Water Tanks	26
7.1 Tank Types	26
7.1.1 Sectional Tanks	26
7.1.2 One piece GRP tanks	26
7.1.3 Underground Tanks	26
7.1.4 Cold Water Service (CWS) Tanks	26
7.1.5 Hot Water Service (HWS) Tanks	26
7.1.6 Sprinkler Tanks	26
7.2 Tank Sizing	27
7.2.1 Tanks in Small Spaces	27
7.3 Tank Inlet	28
7.4 Gravity Discharge from a tank	28
7.5 Solar Hot Water Tank Arrangement	29
7.6 Feed and Expansion Tank	29
8 Domestic Hot Water Loads	30
8.1 Peak Loads	30
8.2 Estimating Annual Hot Water Loads	30

IV Drainage and Rainwater	31
Bibliography	32

Acronyms

ACH Air Changes per Hour.

AHU Air Handling Unit.

CWS Cold Water Service.

HWS Hot Water Service.

VOC Volatile Organic Compounds.

To-Do:

Review this section	8
Figure out how to draw psychrometric processes in $\text{\LaTeX}$	11
Filter Resistance dirty/average/clean etc., coil resistance, PHX vs thermal wheel resistance	20
Put an intro to the tank section here. Also need to decide what is going in this section .	26
See Domestic Water System Calculator - Excel Sheet	27
Include filtering (inc. automatic backwash filter), and valves	28
See notes in OneNote link	29

Part I

Ventilation

Chapter 1

Introduction

Example citation for testing: [1].

This chapter will outline some of the basics of ventilation system design, such as Units, and Ventilation Rates.

1.1 Energy Balances

When performing design calculations on air flow rates, we need to know the energy transfer from different fluids either through coils, mixed air, or other means. The following equations describe the energy balance commonly used within Building Services design.

Equation 1.1, shows the relationship between Energy in Kilo-Watts [kW], Mass Flow Rate in kilograms per second [kg/s], Specific Heat Capacity in Joules/Gram-Kelvin [J K/g], and Temperature Difference in Kelvin [K]. Equations 1.2 through 1.4, shows the dimensional analysis of equation 1.1.

$$Q = m C_p \delta T \quad (1.1)$$

Expressing equation 1.1 as units:

$$\left[\frac{kg}{s} \right] \left[\frac{J}{g \cdot K} \right] \left[\frac{K}{1} \right] \quad (1.2)$$

Equation 1.2 can be reduced by removing Kelvin and grams from both the denominator and numerator:

$$\left[\frac{kg}{s} \right] \left[\frac{J}{g \cdot K} \right] \left[\frac{K}{1} \right] \quad (1.3)$$

This gives equation 1.4, note the definition of Watts [W] as a Joule per second [J/s]:

$$\left[\frac{1000}{s} \right] \left[\frac{J}{1} \right] = \left[\frac{1000}{1} \right] \left[\frac{W}{1} \right] = \left[\frac{kW}{1} \right] \quad (1.4)$$

See Equation 1.5 if you want to use volumetric flow rate [m³/s, or L/s] instead of mass flow rate [kg/s].

1.2 Units

Cubic metres per second [m³/s], litres per second [L/s], kilograms per second [kg/s], air changes per hour [Air Changes per Hour (ACH)] and so on.

- m³/s, is used for larger volumetric air flow rates, there are a 1,000 litres in a cubic meter of air.
- L/s, is used for smaller volumetric air flow rates (Single terminal units/grilles etc), a litre is defined as one cubic decimeter.
- kg/s, is used to describe the mass flow rate of air, and is used when calculating energy balances .
- ACH, is used to describe the turnover of air in a room.

Converting from volumetric flow [m³/s], [L/s] to mass flow [kg/s] requires the density of air [ρ] at the working temperature. Table 1.1 shows the density of air at some common temperatures.

Air Temperature °C	Air Density (ρ) kg/m ³
10	1.246
20	1.204
30	1.164

Table 1.1: Density of air at differing air temperatures

So by using a volumetric ventilation rate, which is commonly used to describe the outdoor air provision per person (See Section 1.4 for more information), we need to include density in Equation 1.1 so that we get sensible answers. Equation 1.1, becomes equation 1.5. (Dimensional analysis follows in Equations 1.6 through 1.8, and 1.9 through 1.11 for L/s).

$$Q = m \rho C_p \delta T \quad (1.5)$$

Following the reduction procedure in section 1.1, we can express equation 1.5 in units leaving equation 1.6:

$$\left[\frac{m^3}{s} \right] \left[\frac{kg}{m^3} \right] \left[\frac{J}{g \cdot K} \right] \left[\frac{K}{1} \right] \quad (1.6)$$

Now removing $[m^3/s]$, $[g]$, and $[K]$ from both the denominator and numerator of equation 1.6:

$$\left[\frac{m^3}{s} \right] \left[\frac{kg}{m^3} \right] \left[\frac{J}{g \cdot K} \right] \left[\frac{K}{1} \right] \quad (1.7)$$

Leaving equation 1.8, again note the definition of Watts $[W]$:

$$\left[\frac{1}{s} \right] \left[\frac{1000}{1} \right] \left[\frac{J}{1} \right] = \left[\frac{1000}{1} \right] \left[\frac{W}{1} \right] = \left[\frac{kW}{1} \right] \quad (1.8)$$

The same procedure can be followed for litres per second $[L/s]$ instead of cubic metres per second $[m^3/s]$, noting the definition of $[L/s]$ as a cubic decimeter, i.e. $1 m^3 = 1,000 l$:

$$\left[\frac{l}{s} \right] \left[\frac{kg}{m^3} \right] \left[\frac{J}{g \cdot K} \right] \left[\frac{K}{1} \right] \quad (1.9)$$

To remove both the litres from the first term numerator, and cubic meters from the second term denominator, we need to divide litres by 1,000, i.e. multiply by $1/1000$.

$$\left[\frac{l}{s} \right] \left[\frac{1}{1000} \right] \left[\frac{kg}{m^3} \right] \left[\frac{J}{g \cdot K} \right] \left[\frac{K}{1} \right] \quad (1.10)$$

$$\left[\frac{1}{s} \right] \left[\frac{J}{1} \right] = \left[\frac{W}{1} \right] \quad (1.11)$$

the primary difference between using $[m^3/s]$ and $[L/s]$ is the addition of $1/1000$ when cancelling the $[m^3]$ and $[L]$; however, the $1/1000$ further cancels out and the end result provides Watts instead of kilo-Watts.

1.3 Shadow Gaps and Door Undercuts

Door undercuts are used as a method of distributing air between rooms without ductwork, by simply allowing space beneath the door for air to pass then sufficient air can be introduced for a small occupied space provided that there is a method of extraction on the other side of

the door. The supply and extraction provides the air flow between the rooms, and the door undercut provides the air path.

Shadow Gaps can provide an air path to a return air plenum, this avoids to use of return air grilles. This can provide a more aesthetically pleasing finish.

The calculation of free area is the same for both door undercuts and shadow gaps. The difference being that the length of the door undercut is determined by the width of the door so only the height is adjustable, whereas the length and width of a shadow gap is adjustable. Equation 1.12, shows the calculation of free area for a given volumetric flow rate, where Q is the volumetric flow rate [m^3/s], v is the velocity of the air stream through the gap [m/s], and A is the free area [m^2].

$$Q = vA \quad (1.12)$$

When using Equation 1.12 to calculate the free area of a door undercut, then by knowing the door width, the height of the undercut can be specified. This two step calculation can be combined into one as shown in equation 1.13, where h is the undercut height [m], and w is the door width [m].

$$h = \frac{Q}{wv} \quad (1.13)$$

Sensible air velocities for shadow gaps and door undercuts are below 2.0 m/s . This avoids excess pressure, and noise generation.

1.4 Ventilation Rates

The amount of air supplied or extracted from a space differs depending on the purpose of ventilation. If air is supplied to maintain a good indoor air quality then the concentration of pollutants (Such as CO_2) in both the supply air and indoor air needs to be considered alongside the generation of pollutants, see subsection 1.4.1. Rules of thumb for the amount of outdoor air that should be provided per person are frequently given in guides such as CIBSE Guide A, some outdoor air rates per person and the expected quality of indoor air are given in Table 1.2.

Supplied Air Rate l/s/p	Indoor Air Quality	Classification [BS EN 13779]	Total Indoor CO ₂ [ppm]
6	Low	IDA4	1600
8	Moderate	IDA3	1200
12.5	Medium	IDA2	900
20	High	IDA1	750

Table 1.2: Commonly Used Outdoor Air Rates, see CIBSE Guide A and BS EN 13779

1.4.1 Pollutant Control

Review this section

An internal space may suffer from a build-up of pollutants such as CO₂, for Volatile Organic Compounds (VOC). The levels of these pollutants within the space can be controlled using mechanical ventilation, by providing external to the space and removing some of the internal air, the concentration of the pollutant can be reduced. Equations 1.14, and 1.15, describe the build-up of a pollutant both over time, and at a steady state condition.

CO₂ is typically released through respiration, an adult human would typically release 0.005 L/s each, and outdoor air typically contains 310 ppm of CO₂ (0.00031 L per L/s of outdoor air).

$$C_t = \frac{q_v c_0 + q_e}{q_v + q_e} \left[1 - e^{-\frac{q_v + q_e}{V} t} \right] \quad (1.14)$$

$$C_E = \frac{q_v c_0 + q_e}{q_v + q_e} \quad (1.15)$$

1.4.2 Kitchen Ventilation Rates

In the DW172 document published by the Heating and Ventilation Contractors Association (HVCA), there are five methods of determining the appropriate extract rate for kitchen extract systems. A mechanical consultant will usually have the extract rate specified by the kitchen specialist, but early stage designs specialists are not usually appointed so an estimation is usually required.

If mechanical supply air is to be provided to the space it is done so at 85% of the extract volumetric flow rate.

The thermal convection method (Method 1) is the only acceptable method for final design, the other four methods should only be used for indicative or provisional extract flow rates.

Item	Area	Fuel	Coefficient	Flow Rate
Griddle	0.4500	Gas	0.30	0.135
Open Top Range	0.6750	Gas	0.35	0.236
Solid Top Range	0.5625	Gas	0.60	0.338
Bench	0.3750	-	0.03	0.011
Twin Fryers	0.4875	Electric	0.35	0.171
Salamander Grille	0.3000	Gas	0.75	0.225
Sub-total Extract Volume				1.116
Canopy Factor - <i>Overhead wall open at both ends</i>				1.25
Volumetric Extract Rate Required				1.395 m ³ /s

Table 1.3: Example method of calculating the extract flow rate using Method 1 from DW172

Loading	Face Velocity
Light (Boiling pans etc.)	0.25 m/s
Medium (Deep fat fryers etc.)	0.35 m/s
Heavy (Chargrills etc.)	0.5 m/s

Table 1.4: Face velocity for the calculation of the extract flow rate using Method 2 from DW172

Extract Sizing - Method 1

This method uses the thermal convection method, and specifies a [m³/s] per [m²] of catering appliance work area. Table 2 in DW172 shows volumetric extract flow rates for different appliances, then applying a canopy factor. An example of calculating the total extract rate is given in Table 1.3.

Extract Sizing - Method 2

If the size of the canopy is known, but not the full detail of the catering equipment being installed, then by using the face velocity method the total extract rate can be estimated. Table 1.4 shows the velocities for different equipment types, and the fastest face velocity should be used.

Extract Sizing - Method 3

A volumetric extract flow rate calculated by assigning a [m³/s] value per [kW] of power input, the total air supply is then totalled.

Extract Sizing - Method 4

Method 4 is based on air changes per hour [ACH], starting at 40 ACH for volumetric extract and increasing this if high output or densely packed equipment is used. ACH rates of up to

120 can be seen, and 40 ACH is regarded as the minimum.

Extract Sizing - Method 5

An extract flow rate is assigned per metre length of extract hood, a table in DW172 has been provided that gives extract rates for different canopy types and duty levels.

Chapter 2

Psychrometrics

Figure out how to draw psychrometric processes in L^AT_EX

Use my [OneNote Psychrometric Notes](#) to help here!

2.1 Psychrometric Charts

2.1.1 Psychrometric Processes

2.1.2 ADP Calculations

2.1.3 Room Condition Calculations

2.1.4 Enthalpy Calculations

2.2 Manual Psychrometric Calculations

Chapter 3

Heating and Cooling via Air

3.1 Heating Requirements

Fabric, Ventilation, and Infiltration

3.2 Cooling Requirements

Fabric, Ventilation, Infiltration, and Internal Gains

3.3 Comfort and Efficiency

Supply air temperatures,

3.4 Calculations

3.4.1 Heating by Air

$$Q = m C_p \delta T$$

3.4.2 Cooling by Air

$$Q = m \delta H$$

Chapter 4

Ventilation Plant

4.1 Air Handling Unit (AHU) Components

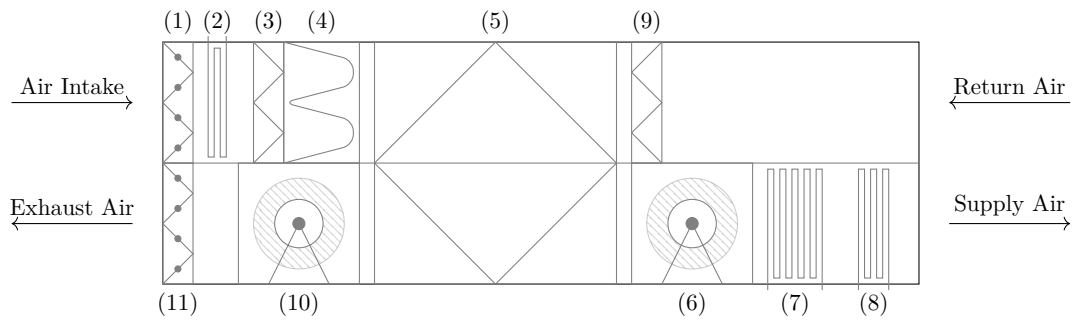


Figure 4.1: Typical AHU schematic, the component numbering follows the air path from outdoor air (Air Intake) through to the space (Supply Air), and then from the return air back to exhaust air.

A typical AHU schematic is shown in figure 4.1, (1) & (11) are motorised dampers, (2) is a frost coil, (3) & (9) are panel filters, (4) is a bag filter, (5) is a counter-flow plate heat exchanger, (6) & (10) are supply and extract fans respectively, (7) is a cooling coil, (8) is a heating or re-heat coil.

4.1.1 Energy Recovery Devices

A number of heat recovery devices are available for inclusion within an AHU.

- Plate Heat Exchanger,
- Thermal Wheel,

- Run-Around-Coil,
- Mixing Box,

Plate Heat Exchanger

The plate heat exchanger works by passing the opposing air streams (supply and extract) through fins within the unit and heat transfer occurs through the metal fins without the two air streams coming into contact with each other. See Figure 4.2 for a schematic example of a plate heat exchanger in a cross flow configuration.

Thermal Wheel

Thermal wheel units utilise a metal perforated wheel which is constantly rotating, this warms up in the extract air stream and then the warm section is rotated into the supply air stream and transfers heat to the air. Thermal wheels mix small amounts of the air streams, and involves the running of a motor to drive the wheel.

Run-Around-Coil

Run-Around-Coil units use two finned coils (similar to the heating and cooling coils) filled with a glycol/water mixture, and run the mixture between the two coils positioned in each of the air streams (supply and extract). This system utilises a pump, and associated pressurisation equipment in order to pass water/glycol between the two coils.

Mixing Box

Mixing Box units recirculate a proportion of the air streams in order to satisfy external air supply and heating/cooling loads simultaneously, for example if the air supply to support the cooling load of a space is double the outdoor air provision, then a mixing box can be used to recirculate half the cooling load air flow rate and allow the rest to come from outside. This can reduce the amount of conditioning required for the supply air as the recirculated air may be closer to the required set-point than the external air.

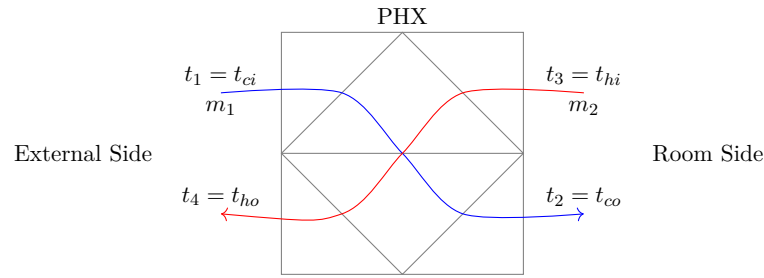


Figure 4.2: A counter-flow plate heat exchanger (PHX) showing air flow paths and temperature labels corresponding to Equation 4.1.

Heat Recovery Type	ErP 2018 Minimum Efficiency
Plate Heat Exchanger	73%
Thermal Wheel	73%
Run-Around-Coil	68%

Table 4.1: Table showing the minimum efficiency required for the three acceptable heat recovery equipment types in air handling units

ErP Regulations

Under the ErP regulations¹, some form of energy recovery must be used and only plate heat exchangers, thermal wheels, or run-around-coils are acceptable. In general the regulations apply to bi-directional and uni-directional ventilation units within commercial sized buildings; however, there are exceptions for certain applications such as Swimming Pools, Agricultural Buildings, Foundries etc.

4.1.2 Energy Recovery Calculations

Equation 4.1 shows the equation describing the efficiency of the heat recovery unit (η) and the temperatures at each point for balanced air flow rates (i.e. supply air flow rate is equal to extract air flow rate), this equation is derived in Derivation 1 and shows allows for unbalanced air streams (i.e. supply air rate does not equal the extract air rate). Figure 4.2 shows a plate heat exchanger (PHX) and the appropriate labels to correspond to Equation 4.1. Minimum values prescribed in the ErP regulations for heat recovery efficiency can be found in Table 4.1.

$$\eta = \frac{t_2 - t_1}{t_3 - t_1} = \frac{t_{co} - t_{ci}}{t_{hi} - t_{ci}} \quad (4.1)$$

¹Known by many different names: Energy related Products (ErP), Ecodesign Directive (№ 2009/125/EC), Lot 6 - Ventilation Units, Ecodesign Regulation (№ 1253/2014), UK Statutory Instrument 2015 (№ 469)

Derivation 1 *This is a derivation of the energy balance within an AHU heat recovery device allowing for unbalanced air flow rates. First we start with the basic energy balance equation defined in Equation 1.1.*

$$Q = m C_p \delta t$$

Now we consider the supply air air-path (Cold), and the extract air air-path (Hot) separately.

$$Q_{\text{cold}} = m C_p (t_{\text{co}} - t_{\text{ci}})$$

$$Q_{\text{hot}} = m C_p (t_{\text{hi}} - t_{\text{ho}})$$

Using the above energy balances we can calculate the cold out temperature (Supply Air, Room side) if we know the temperature of the other three air streams, as this is rarely the case, we need to make some assumptions

If the heat exchanger were to be infinitely long, then the temperature of the hot in and cold out positions would be equal, i.e.

$$t_{\text{hi}} = t_{\text{co}}$$

Now in order to know the air temperature leaving the heat exchanger on the supply air side (t_{co}), we would only have to know what the extract air temperature is (t_{hi}); similarly, the temperature of the hot out (t_{ho}) and cold in (t_{ci}) positions would be equal, i.e.

$$t_{\text{ho}} = t_{\text{ci}}$$

An infinitely long heat exchanger is impractical (and impossible), so we can account for the reduction in heat transfer by using an efficiency term (η) as a ratio of the energy balances, i.e.

$$\frac{Q_{1,\text{cold}}}{Q_{2,\text{hot}}} = \frac{m_1 C_p (t_{\text{co}} - t_{\text{ci}})}{m_2 C_p (t_{\text{hi}} - t_{\text{ho}})}$$

$$\text{let } \eta \times t_{\text{ci}} = t_{\text{ho}}$$

$$\eta = \frac{m_1 \cancel{C_p} (t_{\text{co}} - t_{\text{ci}})}{m_2 \cancel{C_p} (t_{\text{hi}} - t_{\text{ci}})}$$

using the labels in Figure 4.2, the equation becomes;

$$\eta = \frac{m_1 t_2 - t_1}{m_2 t_3 - t_1}$$

QED

4.1.3 Frost Coils

During the winter ventilation design condition, which is typically fully saturated air at temperatures lower than 0 °C, the build-up of ice within an AHU is possible. Ice can build-up on filters, plate-heat exchangers, or coils that are not heated (i.e. cooling coils or in-active heating

coils), but typically only when the unit is running. To combat the ice build up, frost coils can be installed which are sized with sufficient capacity to raise the temperature from the external design condition (say -5°C) to a positive temperature (typically 5°C).

Whilst the unit is operational it is fair to assume that the heating coil and energy recovery devices will be activated and so any damage from ice build up will be limited to the filters (shown at locations (2) and (3) in Figure 4.1), as the energy recovery device will warm the incoming air using the return air. When a filter is frozen there is a risk that the filter will break up, or be pulled through the unit by the action of the fans, which will potentially damage the energy recovery device, or downstream filters. Certain manufacturers will guarantee that their filters can be used in freezing conditions without breaking, therefore negating the need for a frost coil but bringing a component of risk.

Using a frost coil within an AHU will increase the internal static pressure required to be overcome by the fans (Increasing specific fan power (SFP)), reduce the difference in temperature across the energy recovery device (reducing the 'free' energy applied to the supply air stream²), and increase the cost of both the unit and installation.

If a frost coil is omitted from an AHU then additional precautions are required, for example, additional control routines would be built into the control system that would shut dampers if the unit begins to freeze and turn the heating coil to full to attempt to defrost any frozen components. If a cooling coil is present within the unit, then it would need to be sited after the heating coil to prevent any potential freezing within the coil; however, this precludes the ability to dehumidify the air stream should it be necessary/desired.

Calculations with and without Frost Coils

If we consider two AHU with both with a design volumetric air flow rate of $1\text{ m}^3/\text{s}$, a energy recovery device with an efficiency, η , of 73%, and one unit with a frost coil and one without, we can compare the difference in running costs. We are considering the winter time design, so external temperature is -5°C , and the AHU supply temperature set point is 20°C .

Lets consider the LTHW load, Equation 4.5 shows the required LTHW mass flow rate with a frost coil at our design duty, and Equation 4.8 shows the required LTHW mass flow rate without a frost coil.

Firstly the AHU with a frost coil, we consider the energy required to heat $1\text{ m}^3/\text{s}$ of air from

²'free' in the sense of recovered energy that would otherwise be lost to atmosphere without the energy recovery device

-5°C to 5°C :

$$Q = 1 \times 1.2 \times 1.005 \times (5 - -5) = 12.06kW \quad (4.2)$$

Now with a off coil temperature of 5°C from our frost coil and a return air temperature of 20°C we can calculate the temperature gain across our energy recovery device (Using Equation 4.1 described earlier):

$$t_2 = (0.73 \times (20 - 5)) + 5 = 15.95^{\circ}\text{C}, \text{ say } 16^{\circ}\text{C} \quad (4.3)$$

Finally we need the energy required to heat our air stream from 16°C to 20°C :

$$Q = 1 \times 1.2 \times 1.005 \times (20 - 16) = 4.82kW \quad (4.4)$$

So in total we require $12.06 + 4.82 = 16.88kW$ of LTHW input, the mass flow rate of which is:

$$m = \frac{16.88}{4.187 \times (80 - 60)} = 0.20kg/s \quad (4.5)$$

Now we consider the same AHU without the frost coil, ignoring any detriment to frozen filters etc. We know that the air stream is at -5°C and our return air is at 20°C , so the temperature rise across our energy recovery device is:

$$t_2 = (0.73 \times (20 - -5)) + -5 = 13.25^{\circ}\text{C}, \text{ say } 13^{\circ}\text{C} \quad (4.6)$$

This is only three degrees lower than our example with a frost coil, meaning that we recover a lot more heat due to the larger temperature difference, raising the temperature 18°C compared to 6°C with a frost coil. Now we need to calculate the required energy to heat our air stream from 13°C to 20°C :

$$Q = 1 \times 1.2 \times 1.005 \times (20 - 13) = 8.44kW \quad (4.7)$$

The LTHW input to this unit is:

$$m = \frac{8.44}{4.187 \times (80 - 60)} = 0.10kg/s \quad (4.8)$$

Frost Coil	HR Temperature (hi)	Heating Demand	Cost per 24 hours
Yes	16 °C	0.20 kg/s	£11.36
No	13 °C	0.10 kg/s	£5.68
Difference	3 °C	0.10 kg/s	£5.68

Table 4.2: Table showing differences in AHU running costs both with and without a frost coil. The AHU considered has a flow rate of 1 m³/s, a energy recovery device with an efficiency of 73%, and a supply air set point of 20 °C. The external air temperature in this example is −5 °C.

This difference in LTHW mass flow rate is vast, and when considering LTHW generation via natural gas boilers (assuming 90% efficiency) the running cost is as below:

Firstly the unit with frost coils:

$$0.20/0.9 = 0.22kg/s \quad (4.9)$$

$$Q = 0.22 \times 4.187 \times (80 - 60) = 18.42kW \quad (4.10)$$

And assuming a cost of 2.57 p/kWh, the LTHW required by the AHU with frost coils costs £11.36 per 24 hours when the external temperature is −5 °C. The AHU without frost coils:

$$0.10/0.9 = 0.11kg/s \quad (4.11)$$

$$Q = 0.11 \times 4.187 \times (80 - 60) = 9.21kW \quad (4.12)$$

Using the costs above, the LTHW generation for the AHU without frost coils costs £5.68 per 24 hours. Although the cost is relatively small this is the cost of generating LTHW to serve the AHU demand, and the difference can add up when considering the complete winter period where temperatures can be negative for a lot longer than 24 hours! This cost analysis does not consider the increase in fan running cost due to the increased resistance from the frost coil, which affects the running cost year-round whether the frost coil is active or not. Table 4.2 give a summary of the running energy and costs for an AHU with and without a frost coil.

4.1.4 Filtration

4.1.5 Fans

4.1.6 Heating/Cooling Coils

Dehumidification

4.2 Component Resistance

Filter Resistance dirty/average/clean etc., coil resistance, PHX vs thermal wheel resistance

4.3 Fan Heat Gain

4.4 Ventilation Plant Running Costs

Chapter 5

Air Distribution

5.1 Displacement Ventilation

5.2 Ductwork Component Pressure Loss

Part II

Heating and Cooling

Chapter 6

Introduction

6.1 Steady State Heating

There are several steps to calculate the required heating for a room and the size of the central plant that is required, the general procedure is as follows:

- Calculate Conduction (Fabric) Heat Loss,
- Calculate Infiltration Heat Loss,
- Calculate Ventilation System Heat Loss (If Applicable),
- Allow for Local Effects,
- Determine Intermittent Correction Factor,
- Calculate Peak Heating Load,
- Size/Select Heat Emitters,
- Calculate Distribution System Heat Loss,
- Determine Infiltration Diversity Ratio,
- Calculate Central Plant Size,
- Select Central Plant, including any redundancy.

6.1.1 Conduction (Fabric) Heat Loss

$$Q = UA \tag{6.1}$$

Define U-value here...

$$U = \frac{1}{(\sum R + R_i + R_o)} \quad (6.2)$$

Where R is the thermal resistance of the elements, R_i and R_o are the internal and external thermal resistances respectively.

[Designing Buildings Wiki on U-values](#)

6.1.2 Infiltration Heat Loss

e.g. A air pressure test result of $5 \text{ m}^3/(\text{m}^2 \text{ h})$ at 50 Pa would result in a air infiltration rate of 0.111 ACH.

6.1.3 Ventilation Heat Loss

6.1.4 Intermittent Correction Factor

6.1.5 Peak Heating Load

6.1.6 Distribution Losses

6.1.7 Infiltration Diversity Ratio

6.1.8 Central Plant Size

6.2 Annual Estimation of Heating Loads

Part III

Domestic Water Services

Chapter 7

Water Tanks

Put an intro to the tank section here. Also need to decide what is going in this section

7.1 Tank Types

7.1.1 Sectional Tanks

Sectional tanks usually require levelling steels as a base to be installed upon, these can be 50 mm high.

Externally Flanged

Internally Flanged

7.1.2 One piece GRP tanks

7.1.3 Underground Tanks

7.1.4 Cold Water Service (CWS) Tanks

7.1.5 Hot Water Service (HWS) Tanks

7.1.6 Sprinkler Tanks

Sprinkler tanks may need to conform to LPCB requirements. In terms of sizing and specifying a tank, this means that a raised ball-valve chamber is required in order to maintain a class AB air gap.

7.2 Tank Sizing

See Domestic Water System Calculator - Excel Sheet

7.2.1 Tanks in Small Spaces

Sometimes a tank of a certain capacity is required but the space to install a tank is limited. The following subsections detail the usually allowed dimensions and sizes, alongside alternatives that may enable a tank to squeeze into a tight space.

Access Space

Usually 600 mm is allowed as a minimum around the tank in order to tighten the bolts on an externally flanged tank; however, by using a totally internally flanged tank (described in subsection 7.1.1), the access space can be reduced to nothing and the tank installed within the corner of a room. Access is still required to the manway and valve chambers.

Raised Ball Valve Chamber

Typically a raised ball valve chamber can help with space restrictions where the room height is not the same throughout the room; for example, where the ceiling is stepped or sloped. The raised ball valve chamber can be positioned at the higher end of the room allowing the full height of the tank to be used to store water in the lower end of the room. Typical raised ball valve chambers are 450 mm high, whereas manufacturers sometimes offer smaller 'special' raised ball valve chambers such as a 250 mm high chamber from Balmoral Tanks which has side access to the valve.

Manway

Access to the internals of the tank is required for periodic cleaning when the tank is drained. Standard arrangements involve manways located on the top surface of the tank, complete with a fixed 'cat' ladder. Where height is restricted, the manway can be relocated to the side wall of the tank, with a standard ladder on both the inside and outside of the tank to allow access, one drawback is that the tank has to be drained before the manway can be opened.

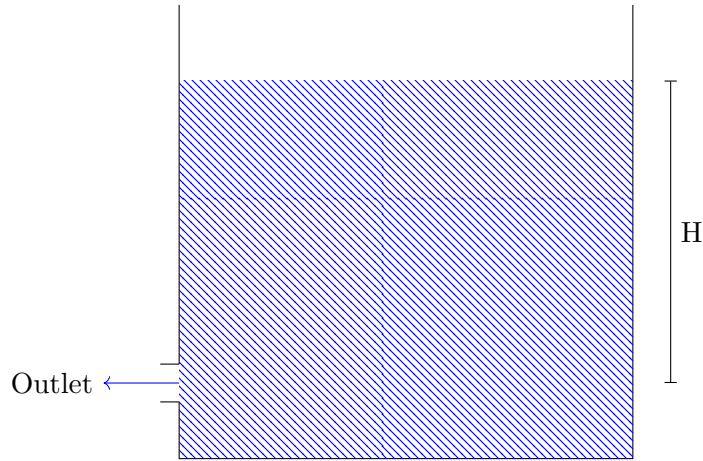


Figure 7.1: Sidewall gravity discharge from a tank

7.3 Tank Inlet

Include filtering (inc. automatic backwash filter), and valves

7.4 Gravity Discharge from a tank

Discharge from a tank by gravity alone is dependant on the following; Outlet Area (A), Acceleration due to Gravity (g), Height from the outlet to the water level (H), and the Coefficient of Discharge (C_d). Figure 7.1 shows a tank open to atmosphere with a side-wall discharge, and Equation 7.1 describes the volumetric flow rate through the outlet based on gravity flow, and an example tank discharge has been calculated in Equation 7.2.

$$Q = C_d A (2gH)^{\frac{1}{2}} \quad (7.1)$$

$$Q = 0.5 \times 0.00008 \times (2 \times 9.81 \times 0.8)^{0.5} = 0.00016 \text{ m}^3/\text{s} = 0.16 \text{ litre/s} \quad (7.2)$$

To calculate the centre line velocity of the discharge (U), Equation 7.3 is used, and example for the same tank has been shown in Equation 7.4. C_v has been assumed to be 0.97 in this example.

$$U = C_v (2gH)^{\frac{1}{2}} \quad (7.3)$$

$$U = 0.97 \times (2 \times 9.81 \times 0.8)^{0.5} = 3.84 \text{ m/s} \quad (7.4)$$

Finally the reaction force (F) from the water nozzle can be calculated from Equation 7.5, the density of water (ρ) has been taken as 1.000 kg/m^3 . The example tank has been shown in Equation 7.6

$$F = \rho QU \quad (7.5)$$

$$F = 1,000 \times 0.00016 \times 3.84 = 0.61 \text{ kg m/s}^2 = 0.61 \text{ N} \quad (7.6)$$

7.5 Solar Hot Water Tank Arrangement

See notes in OneNote link

[OneNote Solar HW Integration Notes](#)

7.6 Feed and Expansion Tank

Chapter 8

Domestic Hot Water Loads

8.1 Peak Loads

8.2 Estimating Annual Hot Water Loads

Part IV

Drainage and Rainwater

Bibliography

- [1] Adam Rees, Lydia C Powell, Gary Chinga-Carrasco, David T Gethin, Kristin Syverud, Katja E Hill, and David W Thomas. 3D Bioprinting of Carboxymethylated-Periodate Oxidized Nanocellulose Constructs for Wound Dressing Applications. *BioMed Research International*, 2015:1–7, 2015.